7:04 4G 69%
quiz.vu.edu.pk/QuizQuestio 11

MTH622:Quiz no.4 Quiz Start Time: 07:01 PM, 10

Question # 5 of 5 ( Start time: 07:04:20 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{yy}$

Select the correct option

Reload Ma

- 7m
- 8m ✓

Click to Save Answer & Move to Next Que

$I_{xx} = 7m$

$I_{yy} = 8m$

$I_{zz} = 9m$

LMS LMS

7:04 4G 69%
quiz.vu.edu.pk/QuizQuestio 11

MTH622:Quiz no.4 Quiz Start Time: 07:01 PM, 10

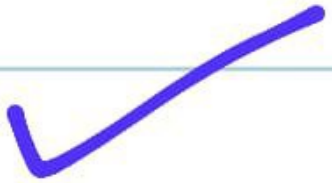
Question # 5 of 5 ( Start time: 07:04:20 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{yy}$

Select the correct option

Reload

- 7m
- 8m



Click to Save Answer & Move to Next Que

6:46 86%
quiz.vu.edu.pk/QuizQue
6

MC210200644: AROOJ KARIM

Time Left 88 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 06:46 PM, 10 March 2022

Question # 1 of 5 ( Start time: 06:46:37 PM, 10 March 2022 )

Total Marks: 1

The moment of inertia of a plane rigid body about an axis perpendicular to the body is equal to the difference of the moment
of inertia about two mutually perpendicular axis lying in the plane

Select the correct option

- false
- true

Click to Save Answer & Move to Next Question

Sum
True

LMS LMS LMS LMS

Question # 1 of 5 ( Start time: 06:50:51 PM, 10 March 2022 )

Point, where the total volume of the body is assumed to be concentrated is----.

Select the correct option

- center of area
- all of the mentioned
- centroid of mass
- centroid of volume

ston # 3 of 5 ( Start time: 06:52:33 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{yy}$

Select the correct option

- $7m$
- $3m$

$I_{yy}$

6:48 85%
quiz.vu.edu.pk/QuizQue
6

MC210200644: AROOJ KARIM

Time Left 86 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 06:46 PM, 10 March 2022

Question # 4 of 5 ( Start time: 06:48:29 PM, 10 March 2022 )

Total Marks: 1

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

Reload Math Equations

- coincident
- perpendicular

Click to Save Answer & Move to Next Question

6:49 85%
quiz.vu.edu.pk/QuizQue
6

MC210200644: AROOJ KARIM

Time Left 87 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 06:46 PM, 10 March 2022

Question # 5 of 5 ( Start time: 06:49:06 PM, 10 March 2022 )

Total Marks: 1

The axis about which moment of area is taken is known as ——.

Select the correct option

Reload Math Equations

- axis of reference
- axis of area
- axis of rotation
- axis of moment

Click to Save Answer & Move to Next Question

6:47 86%
quiz.vu.edu.pk/QuizQue

MC210200644: AROOJ KARIM

Time Left 88 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 06:46 PM, 10 March 2022

Question # 2 of 5 ( Start time: 06:47:13 PM, 10 March 2022 )

Total Marks: 1

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate systems, then $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$ where it is not necessary that the origins of the two systems coincide initially.

Select the correct option

Reload Math Equations

- true
- false

Click to Save Answer & Move to Next Question

principal

## Related Theorems

### ➢ Theorem 1

A $3 \times 3$ symmetrical matrix has three real eigenvalues, which may be distinct or repeated.

### ➢ Theorem 2

The eigenvectors of a symmetrical matrix corresponding to distinct eigenvalues are orthogonal.

### ➢ Theorem 3

It is always possible to find three mutually orthogonal eigenvectors for a $3 \times 3$ symmetric matrix, whether the eigenvalues are distinct or repeated.

## Results

In the light of above theorems, we deduce the following results about the inertia matrix $I_{mat}$ at point $O$ of a rigid body.

i. The principal of moment of inertia are always real numbers. This is obviously physical because the moment of inertia is defined as the quantity $\sum_i m_i d_i^2$ where $m_i$ and $d_i$ are both real.

3:22

Quiz
https://quiz.vu.edu.pk

MC210203829: ZAIB UN NISA

Time Left 88 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 03:19 PM, 10 March 2022

Question # 3 of 5 ( Start time: 03:21:59 PM, 10 March 2022 )

Total Marks: 1

Three particles of masses $m, 2m$ and $3m$ are held in a rigid light framework at points $(0,1,1), (1,0,1)$ and $(1,1,0)$, respectively. Find $I_{xx}$

Select the correct option

Reload Math Equations

- 8m
- 7m

Click to Save Answer & Move to Next Question

6:47 85%
quiz.vu.edu.pk/QuizQue
6

MC210200644: AROOJ KARIM

Time Left 87 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 06:46 PM, 10 March 2022

Question # 3 of 5 ( Start time: 06:47:50 PM, 10 March 2022 )

Total Marks: 1

Perpendicular axis theorem can be applied for which of the following bodies?

Select the correct option

Reload Math Equations

- A cube of side 'a'
- Cylinder of radius R and height h
- Disc of radius R and thickness t
- Ring having radius R & negligible cross section

Click to Save Answer & Move to Next Question

Module No. 17

# Perpendicular Axis Theorem

## Theorem Statement

The moment of inertia of a **plane rigid body** about an axis perpendicular to the body is equal to the sum of the moment of inertia about two mutually perpendicular axis lying in the plane of the body and meeting at the common point with the given axis.

# Example 5 of Moment of Inertia

the parallel axis theorem

## Theorem Statement

The moment of inertia of a rigid body about a given axis is equal to the same about a parallel axis through the centroid plus the moment of inertia due to the total mass placed at the centroid (the last quantity will be referred to as moment of inertia of the centroid).

## Proof

...ition, if $I$ denotes the M.I about the ...

$$M = \sum_i m_i$$

is the total mass of the system.

or

$$I = I_0 + I'$$

where $I_0$ denotes the moment of inertia of the centroid, and $I'$ denotes the M.I of the syster w.r.to a parallel axis through the centroid.

MTH622:Quiz no.4

Quiz Start Time: 04:44 PM, 10 March 2022

Question # 4 of 5 ( Start time: 04:47:23 PM, 10 March 2022 )

Total Marks: 1

Three particles of masses $m, 2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1), (1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{xx}$

Select the correct option

Reload Math Equations

- 7m
- 8m

Click to Save Answer & Move to Next Question

MTH622

#160

MTH622:Quiz no.4

Quiz Start Time: 04:44 PM, 10 March 2022

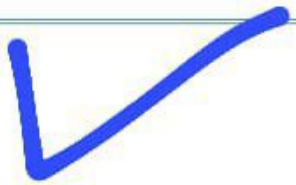
Question # 2 of 5 ( Start time: 04:45:51 PM, 10 March 2022 )

Total Marks: 1

The moment of inertia of a rigid body about a given axis is equal to the same about a parallel axis through the centroid plus
the moment of inertia due to the total mass placed at the centroid

Select the correct option

- false
- true



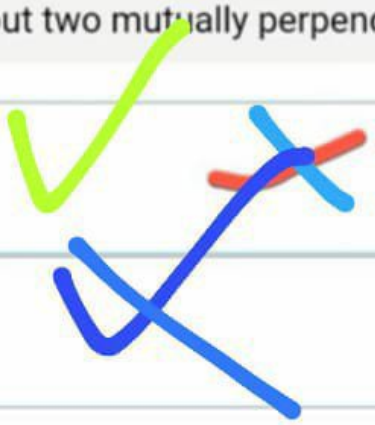
Click to Save Answer & Move to Next Question

Question # 2 of 5 ( Start time: 03:20:47 PM, 10 March 2022 ) Total Marks: 1

The moment of inertia of a plane rigid body about an axis perpendicular to the body is equal to the difference of the moment
of inertia about two mutually perpendicular axis lying in the plane

Select the correct option

- false
- true



Question # 2 of 5 ( Start time: 04:50:38 PM, 10 March 2022 ) Total Marks: 1

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate systems, then $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$ where it is not necessary that the origins of the two systems coincide initially.

Select the correct option

Reload Math Equations

- false
- true

Click to Save Answer & Move to Next Question

MTH622:Quiz no.4

Quiz Start Time: 04:44 PM, 10 March 2022

Question # 5 of 5 ( Start time: 04:48:04 PM, 10 March 2022 )

Total Marks: 1

Given that the moment of inertia of the disc about a line passing through the center and perpendicular to the plane of disc is $\frac{1}{2}Mr^2$, then the moment of inertia of the disc about any diameter is

Select the correct option

Reload Math Equations

- $\frac{1}{4}Mr^2$ ✓
- $\frac{1}{2}Mr^2$

Click to Save Answer & Move to Next Question

module

#171

MTH622:Quiz no.4

Quiz Start Time: 04:44 PM, 10 March 2022

Question # 4 of 5 ( Start time: 04:47:23 PM, 10 March 2022 )

Total Marks: 1

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{xx}$

Select the correct option

Reload Math Equations

- 7m
- 8m

Click to Save Answer & Move to Next Question

3:58 PM 13.0 K/s 82

quiz.vu.edu.pk

Question # 3 of 5 ( Start time: 03:58:36 PM, 10 March 2022 )

Three mutually orthogonal eigenvectors for a 3×3 symmetric matrix exist if

Select the correct option

- the eigenvalues are either distinct or repeated
- the eigenvalues are only distinct
- the eigenvalues are only repeated
- the eigenvalues are imaginary

Click to Save Answer

Search...

3:56 PM

quiz.vu.edu.pk

TH622:Quiz no.4 Quiz Start Time: 03:56 PM, 1

estion # 1 of 5 ( Start time: 03:56:38 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0,1,1)$, $(1,0,1)$ and $(1,1,0)$, respectively. Find $I_{xx}$

lect the correct option

Reload M

- 8m
- 7m

Click to Save Answer & Move to Next Q

Search...

MTH622:Quiz no.4

Quiz Start Time: 04:50 PM, 10 March 2022

Question # 5 of 5 ( Start time: 04:52:58 PM, 10 March 2022 )

Total Marks: 1

Perpendicular axis theorem can be applied for which of the following bodies?

Select the correct option

Reload Math Equations

- Disc of radius R and thickness t
- A cube of side 'a'
- Ring having radius R & negligible cross section
- Cylinder of radius R and height h

Click to Save Answer & Move to Next Question

MTH622:Quiz no.4

Quiz Start Time: 04:44 PM, 10 March 2022

Question # 5 of 5 ( Start time: 04:48:04 PM, 10 March 2022 )

Total Marks: 1

Given that the moment of inertia of the disc about a line passing through the center and perpendicular to the plane of disc is $\frac{1}{2}Mr^2$, then the moment of inertia of the disc about any diameter is

Select the correct option

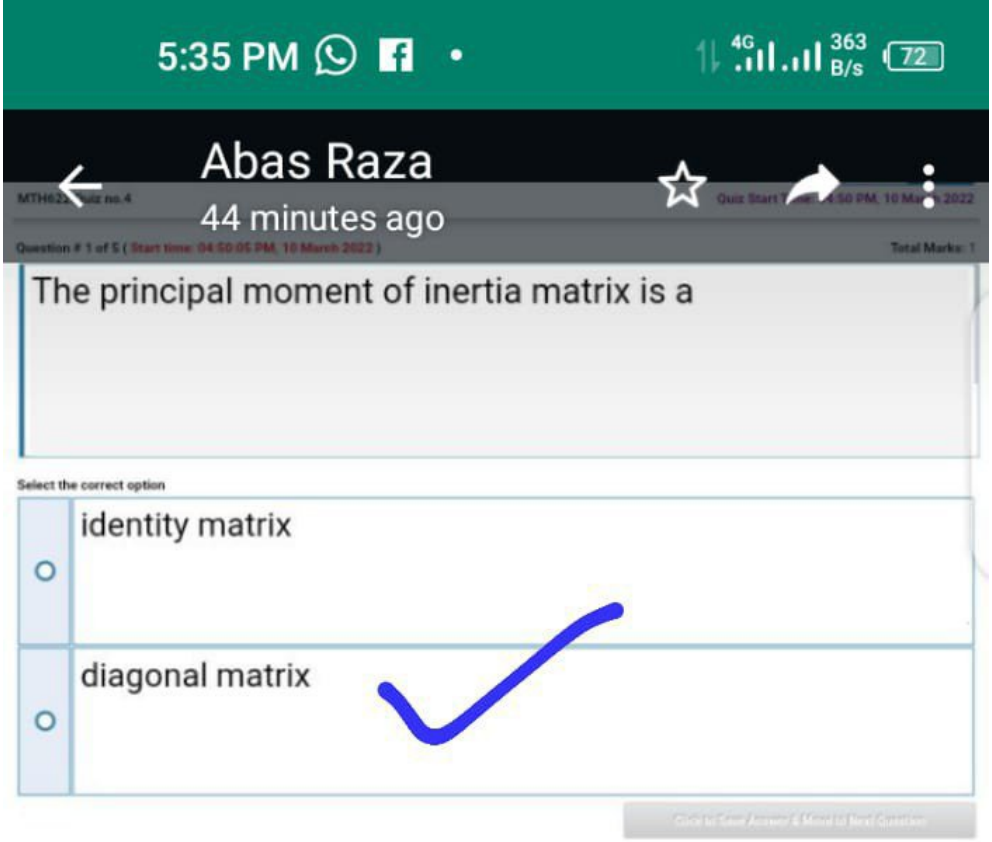
Reload Math Equations

- $\frac{1}{4}Mr^2$ ✓
- $\frac{1}{2}Mr^2$

Click to Save Answer & Move to Next Question

module

#171


5:35 PM
363 B/s
72
Abas Raza
44 minutes ago
The principal moment of inertia matrix is a
Select the correct option
identity matrix
diagonal matrix

#177

MTH622:Quiz no.4

Quiz Start Time: 04:44 PM, 10 March 2022

Question # 4 of 5 ( Start time: 04:47:23 PM, 10 March 2022 )

Total Marks: 1

Three particles of masses $m, 2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1), (1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{xx}$

Select the correct option

Reload Math Equations

- 7m
- 8m

Click to Save Answer & Move to Next Question

MTH622

#160

5:02 PM 549 B/s 69

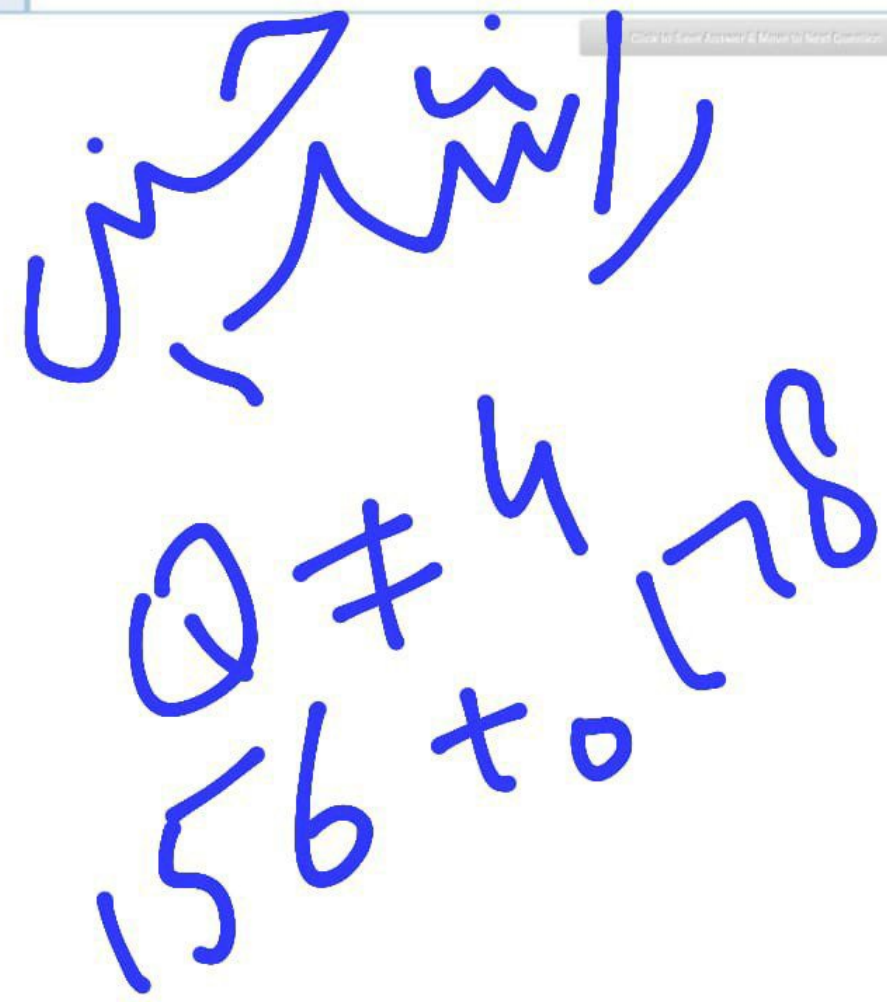
Abas Raza
10 minutes ago

Question # 4 of 5 ( Start time: 04:52:26 PM, 10 March 2022 ) Total Marks: 1

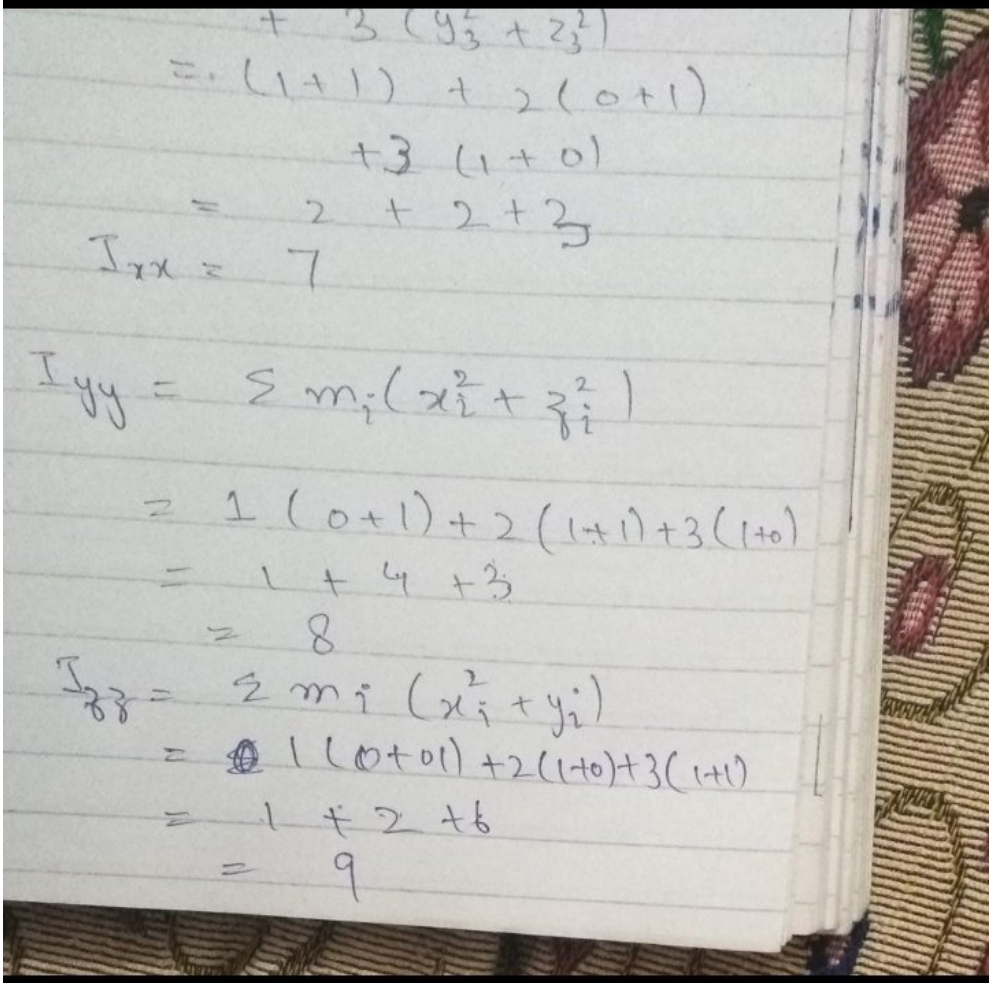
The moment of inertia of a rigid body about a given axis is equal to the same about a parallel axis through the centroid plus
the moment of inertia due to the total mass placed at the centroid

Select the correct option

- true
- false



5:04 PM
Malik Aftab Vu
11 minutes ago



$+ 3(y_3^2 + z_3^2)$

$= (1+1) + 2(0+1)$

$+3(1+0)$

$= 2 + 2 + 3$

$I_{xx} = 7$

$I_{yy} = \sum m_i (x_i^2 + z_i^2)$

$= 1(0+1) + 2(1+1) + 3(1+0)$

$= 1 + 4 + 3$

$= 8$

$I_{zz} = \sum m_i (x_i^2 + y_i^2)$

$= 1(0+0) + 2(1+0) + 3(1+1)$

$= 1 + 2 + 6$

$= 9$

6:14 PM • 915 B/s 69

You
Just now

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MTH622:Quiz no.4 Quiz Start Time: 04:13 PM

Question # 2 of 5 ( Start time: 04:13:45 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{zz}$

Select the correct option

Reload

- 9m
- 8m

Click to Save Answer & Move to Next

6:14 PM • 915 B/s 69

You
Just now
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MTH622 Quiz no 4 — Quiz Start Time: 04:13 PM

Question # 2 of 5 ( Start time: 04:13:45 PM, 10 March 2022 )

Three particles of masses m, 2m and 3m are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{xx}$

Select the correct option

- 9m
- 8m

Click to Save Answer & Move to Next

$$r_i = r_c + r_i'$$

On making substitutions, we obtain

$$I = \sum_i m_i \left[e \times (r_c + r_i')\right]^2$$

$$= \sum_i m_i \left[e \times r_c + e \times r_i'\right]^2$$

$$= \sum_i m_i \left[(e \times r_c)^2 + (e \times r_i')^2 + 2\left(e \times r_c\right).(e \times r_i')\right]$$

$$= \sum_i m_i (e \times r_c)^2 + \sum_i m_i (e \times r_i')^2$$

$$+2\left(e \times r_c\right).\sum_i m_i (e \times r_i')$$

$$= \sum_i m_i {d_c}^2 + \sum_i m_i {d_i'}^2 + 2\left(e \times r_c\right).e \times \sum_i m_i r_i'$$

But $\sum_i m_i r_i' = 0$ (We studied in earlier module).

Therefore

$$I = (\sum_i m_i){d_c}^2 + \sum_i m_i {d_i'}^2$$

$$= M{d_c}^2 + I'$$

where

$$M = \sum_i m_i$$

is the total mass of the system.

or

$$I = I_0 + I'$$

where $I_0$ denotes the moment of inertia of the centroid, and $I'$ denotes the M.I of the system w.r.to a parallel axis through the centroid.

Equation (3) is a system of three homogeneous linear algebraic equation in the unknown $\omega_1, \omega_2, \omega_3$.

This system will have a non-trivial solution, i.e. ($\vec{\omega} \neq 0$), if the determinant of the matrix of coefficients is zero. i.e.

$$\begin{vmatrix} I_{11} - I & I_{12} & I_{13} \\ I_{21} & I_{22} - I & I_{23} \\ I_{31} & I_{32} & I_{33} - I \end{vmatrix} = 0 \qquad (4)$$

This is a cubic equation in $I$ and will in general have three roots. Equation (4) is called characteristic equation of the matrix $(I_{ij})$.

The roots of equation (4) are called eigen values of the inertia matrix $(I_{ij})$.

The problem of finding principal moment of inertia and directions of inertia has been reduced to that of finding the eigenvalues and eigenvectors of a symmetric $3 \times 3$ matrix.

The following results from the eigenvalue theory of matrices will deduce further results about the principal moments and directions (axes) of inertia.

## Related Theorems

### ➢ Theorem 1

A $3 \times 3$ symmetrical matrix has three real eigenvalues, which may be distinct or repeated.

### ➢ Theorem 2

The eigenvectors of a symmetrical matrix corresponding to distinct eigenvalues are orthogonal.

### ➢ Theorem 3

It is always possible to find three mutually orthogonal eigenvectors for a $3 \times 3$ symmetric matrix, whether the eigenvalues are distinct or repeated.

## Results

In the light of above theorems, we deduce the following results about the inertia matrix $I_{mat}$ at point $O$ of a rigid body.

i. The principal of moment of inertia are always real numbers. This is obviously physical because the moment of inertia is defined as the quantity $\sum_i m_i d_i^2$ where $m_i$ and $d_i$ are both real.

Module No. 17

# Perpendicular Axis Theorem

## Theorem Statement

The moment of inertia of a **plane rigid body** about an axis perpendicular to the body is equal to the sum of the moment of inertia about two mutually perpendicular axis lying in the plane of the body and meeting at the common point with the given axis.

# Example 5 of Moment of Inertia

the parallel axis theorem

## Theorem Statement

The moment of inertia of a rigid body about a given axis is equal to the same about a parallel axis through the centroid plus the moment of inertia due to the total mass placed at the centroid (the last quantity will be referred to as moment of inertia of the centroid).

## Proof

...ition, if $I$ denotes the M.I about the ...

$$M = \sum_i m_i$$

is the total mass of the system.

or

$$I = I_0 + I'$$

where $I_0$ denotes the moment of inertia of the centroid, and $I'$ denotes the M.I of the syster w.r.to a parallel axis through the centroid.

principal

## Related Theorems

### ➢ Theorem 1

A $3 \times 3$ symmetrical matrix has three real eigenvalues, which may be distinct or repeated.

### ➢ Theorem 2

The eigenvectors of a symmetrical matrix corresponding to distinct eigenvalues are orthogonal.

### ➢ Theorem 3

It is always possible to find three mutually orthogonal eigenvectors for a $3 \times 3$ symmetric matrix, whether the eigenvalues are distinct or repeated.

## Results

In the light of above theorems, we deduce the following results about the inertia matrix $I_{mat}$ at point $O$ of a rigid body.

i. The principal of moment of inertia are always real numbers. This is obviously physical because the moment of inertia is defined as the quantity $\sum_i m_i d_i^2$ where $m_i$ and $d_i$ are both real.

Module No. 178

# Relation of Fixed and Rotating Reference

In order to derive the relationship between fixed and rotating frames of the following theorem[1,2].

## Rotating Axes Theorem

### ➤ Theorem Statement

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fix system, then

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$$

where it is understood that the origins of the two systems coincide at $t$ =

### ➤ Proof

We denote the fixed and rotating coordinate systems by $OX_0Y_0Z_0$ and $O$ ssociated unit vectors by $\{i_0, j_0, k_0\}$ and $\{i, j, k\}$.

onsider a vector $A$ which is changing with time. To an observer fixed e time rate of change of $A = A_1\hat{\imath} + A_2\hat{\jmath} + A_3\hat{k}$ will be

$$\frac{d}{dt}A_r = \frac{d}{dt}A_1\hat{\imath} + \frac{d}{dt}A_2\hat{\jmath} + \frac{d}{dt}A_3\hat{k}$$

ere $\frac{dA_r}{dt}$ denotes the time derivative of $A$ relative to the rotating frame

vever, the time rate of change of $A$ relative to the fixed system $OX_0Y$ needs to be found.

ne fixed observer the unit vectors $i$, $j$, $k$ actually change with time.

6:27 PM • 1.6 K/s 67

You
11 minutes ago

MC210203259: ALI HAIDER

MTH622:Quiz no.4

Question # 5 of 5 ( Start time: 04:16:58 PM, 10 March 2022 )

The axis about which moment of area is taken is known as --------.

Select the correct option

- axis of reference
- axis of moment
- axis of rotation
- axis of area

6:27 PM • 1.6 K/s 67
You
11 minutes ago

MC210203259: ALI HAIDER

MTH622:Quiz no.4

Question # 5 of 5 ( Start time: 04:16:58 PM, 10 March 2022 )

The axis about which moment of area is taken is known as --------.

Select the correct option

- axis of reference
- axis of moment
- axis of rotation
- axis of area

4:15 PM
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2

MC210203259: ALI HAIDER

MTH622:Quiz no.4

Question # 3 of 5 ( Start time: 04:15:01 PM, 10 March 2022 )

Perpendicular axis theorem can be applied for which of the following bodies?

Select the correct option

- A cube of side 'a'
- Ring having radius R & negligible cross section
- Disc of radius R and thickness t
- Cylinder of radius R and height h

4:14 PM

quiz.vu.edu.pk/QuizQue:

MC210203259: ALI HAIDER

MTH622:Quiz no.4

Question # 2 of 5 ( Start time: 04:13:45 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1,$

Select the correct option

- 9m
- 8m

2:58 73%
[Solved] The axis a...
mcqmate.com
Search mcq/topic/co

**Q.** The axis about which moment of area is taken is known as .

**A.** Axis of area

**B.** Axis of moment

**C.** Axis of reference

**D.** Axis of rotation

Answer» **C.** Axis of reference

Report

3:20

Quiz
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MC210203829: ZAIB UN NISA

Time Left 87 sec(s)

MTH622:Quiz no.4

Quiz Start Time: 03:19 PM, 10 March 2022

Question # 2 of 5 ( Start time: 03:20:47 PM, 10 March 2022 )

Total Marks: 1

The moment of inertia of a plane rigid body about an axis perpendicular to the body is equal to the difference of the moment
of inertia about two mutually perpendicular axis lying in the plane

Select the correct option

- false
- true

Click to Save Answer & Move to Next Question

True option

Sum

Module No. 178

# Relation of Fixed and Rotating Frames of Reference

In order to derive the relationship between fixed and rotating frames of reference, we will study the following theorem[1,2].

## Rotating Axes Theorem

### ➤ Theorem Statement

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, then

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r \quad \text{or} \quad V_f = V_r + \omega \times r$$

where it is understood that the origins of the two systems coincide at $t = 0$

### ➤ Proof

We denote the fixed and rotating coordinate systems by $OX_0Y_0Z_0$ and $OXYZ$ and denote the associated unit vectors by $\{i_0, j_0, k_0\}$ and $\{i, j, k\}$.

Consider a vector $A$ which is changing with time. To an observer fixed relative to $OXYZ$ system, the time rate of change of $A = A_1\hat{\imath} + A_2\hat{\jmath} + A_3\hat{k}$ will be

$$\frac{d}{dt}A_r = \frac{d}{dt}A_1\hat{\imath} + \frac{d}{dt}A_2\hat{\jmath} + \frac{d}{dt}A_3\hat{k}$$

where $\frac{dA_r}{dt}$ denotes the time derivative of $A$ relative to the rotating frame of reference.

However, the time rate of change of $A$ relative to the fixed system $OX_0Y_0Z_0$ symbolized by the needs to be found.

The fixed observer the unit vectors $i$, $j$, $k$ actually change with time.

7:04 4G 69%
quiz.vu.edu.pk/QuizQuestio 11

MTH622:Quiz no.4 Quiz Start Time: 07:01 PM, 10

Question # 5 of 5 ( Start time: 07:04:20 PM, 10 March 2022 )

Three particles of masses $m$, $2m$ and $3m$ are held in a rigid light framework at points $(0, 1, 1)$, $(1, 0, 1)$ and $(1, 1, 0)$, respectively. Find $I_{yy}$

Select the correct option

- 7m
- 8m ✓

Click to Save Answer & Move to Next Que

$I_{xx} = 7m$

$I_{yy} = 8m$

$I_{zz} = 9m$

LMS LMS